

Search Keywords: Quantum key distribution simulation, component based, program QKD, Fiber-based photonics simulation, advanced optical simulations, component-based QKD, Quantum Simulation Environments, Quantum Network Simulations								
QKD and Quantum Network Simulation Environments over the past 22 years.								
Databases Used: Web of Science, Scopus, Individual Google Scholar Searches, IEEE Xplore								
S.No	Paper Name	Journal/Conference	Year	Author Name(s)	Include/Exclude	Comments	Journal/Conference Name	BibText Key
1	QuCCs: an experimental of quantum key distribution using quantum cryptography and communication simulator	Journal	2019	Zukarnain, ZA and Buhari, A and Harun, NZ and Khalid, R	Include	A QKD simulation environment designed using Java, MATLAB and SQL. Models are empirical and are mostly modeled using distributions. Is not solely for QKD.	Malaysian Journal of Mathematical Sciences	zukarnain2019quccs
2	qkdSim, a Simulation Toolkit for Quantum Key Distribution Including Imperfections: Performance Analysis and Demonstration of the B92 Protocol Using Heralded Photons	Journal	2020	Chatterjee, Rishab and Joarder, Kaushik and Chatterjee, Sourav and Sanders, Barry C. and Sinha, Urbasi	Include	A QKD Simulation environment using components. Not stated if components are empirical or based on proper mathematical models. Does state that the paper presents a software product.	American Physical Society Physical Review Applied	PhysRevApplied
3	Modeling Quantum Optical Components, Pulses and Fiber Channels Using OMNeT++	Journal	2015	Ryan D. L. Engle and Douglas D. Hodson and Michael R. Grimaila and Logan O. Mailloux and Colin V. McLaughlin and Gerald Baumgartner	Include	A QKD Simulation environment produced using OMNeT++, which is part of a modular component based C++ library. Laser is modeled using ID300 approximated via three gaussian pulse shapes. Fiber channel is abstracted, with features added for attenuation, drifts. Does not state information about other models, but an assumption is made that they are also empirically formulated/approximated.	arXiv The Computing Research Repository	EngleHGMMB15
4	EnQuad: A Publicly-Available Simulator for Quantum Key Distribution Protocols	Journal	2020	Abdelgawad, Mohamed S. and Shenouda, Botrous A. and Abdullatif, Sameh O.	Include	A QKD simulation environment produced using MATLAB. This environment has additional features boasting faster runtime and has security testing. Models are not component based, and are modeled using probabilities	Walter de Gruyter Cybernetics and Information Technology	cait-2020-0002
5	SeQUeNCe: a customizable discrete-event simulator of quantum networks	Journal	2021	Xiaoliang Wu and Alexander Kolar and Joaquin Chung and Dong Jin and Tian Zhong and Rajkumar Kettimuthu and Martin Suchara	Include	A QKD simulation environment using Python. Dubbed as Simulator of QUantum Network Communication, which uses a non rigid structure layer (modifieable). Information of hardware level base layer components can be acquired from the authors previous papers, which also use probabilties along with qubit state representation.	IOP Quantum Science and Technology	Wu_2021
6	NetSquid, a NETwork Simulator for QUantum Information using Discrete events	Journal	2020	Coopmans, Tim and Knegjens, Robert and Dahlberg, Axel and Maier, David and Nijsten, Loek and de Oliveira Filho, Julio and Papendrecht, Martijn and Rabbie, Julian and Rozpędek, Filip and Skrzypczyk, Matthew and Wubben, Leon and de Jong, Walter and Podareanu, Damian and Torres-Knoop, Ariana and Elkouss, David and Wehner, Stephanie	Maybe	A QKD simulation environment using Python. It is used to investigate/simulate large quantum network models, and does quantum state tracking via managing state space sizes throughout the model. At its core it has no physical layer and works on node and API's designed to mimic qubit and quantum network behaviors	Springer Communications Physics	Coopmans_2021
7	SimulaQron - A simulator for developing quantum internet software	Journal	2018	Dahlberg, Axel and Wehner, Stephanie	Maybe	A quantum network simulator with provided integration with C and Python. It is used to simulate node-based quantum networks over a quantum channel.	IOP Quantum Science and Technology	Dahlberg_2018
8	A distributed simulation framework for quantum networks and channels	Journal	2018	Ben Bartlett	Include	Dubbed as SQUANCH, is used to design complex multi-party quantum networks, and has extensible classes for modeling noisy quantum channels.	arXiv	bartlett2018
9	QuNetSim: A Software Framework for Quantum Networks	Journal	2020	Diadamo, Stephen and Notzel, Janis and Zanger, Benjamin and Bese, Mehmet Mert	Maybe	A Quantum Network Simulator built on python that simulates quantum networks at network layer and allows for investigation of various quantum networks various network configs and parameters. It also incorporates various well known quantum backends like QuTIP, CQC, ProjectQ. However nothing in this simulator is on the physical layer	IEEE Transactions on Quantum Engineering	Diadamo_2021
10	Implementation of quantum key distribution network simulation module in the network simulator NS-3	Journal	2017	Mehic, Miralem and Maurhart, Oliver and Rass, Stefan and Voznak, Miroslav	Include	A QKD simulation environment designed using C++ as a module for a discrete event network simulator called NS-3. Also called as QKDNetSim. It follows purely emprirical models of qubit and hilbert spaces and is solely for QKD transmission protocols. It handles receiving, transmitting, nodes, network, cryptography handling, key buffers and an imitation quantum channel in form of nodes	Springer Quantum Information Processing	Mehic2017
11	Emulation of Quantum Key Distribution Networks	Journal	2024	Mehic, Miralem and Dervisevic, Emir and Burdiak, Patrik and Lipovac, Vlatko and Fazio, Peppino and Voznak, Miroslav	Include	Emulation of QKD Networks with web interface using QKDNetSim by original authors of QKDNetSim	IEEE Network	QKDNetSim
12	QKDNetSim+: Improvement of the quantum network simulator for NS-3	Journal	2024	Soler, David and Cillero, Iván and Dafonte, Carlos and Fernández-Veiga, Manuel and Vilas, Ana Fernández and Nóvoa, Francisco J.	Include	Extension of QKDNetSim by new authors. Recoded how transmission protocol behaves, adding key storage to key buffers, encryption during cryptography handling	Elsevier SoftwareX	Soler_2024
13	A simulator of optical coherent-state evolution in quantum key distribution systems	Journal	2022	Caputo, Carlo and Simoni, Mario and Cirillo, Giovanni Amedeo and Turvani, Giovanna and Zamboni, Maurizio	Include	Component based model using MATLAB. Relations in this model follow empirical modeling. Coherent source is modeled as a coherent state (Fock States), propogation and most components, including the quantum channel are modeled using only Jones Calculus. However it does follow the Qubit state much closely compared to other component based architectures	Springer Optical and Quantum Electronics	Caputo2022
14	NuQKD: A Modular Quantum Key Distribution Simulation Framework for Engineering Applications	Journal	2024	Gkouliaras, Konstantinos and Theos, Vasileios and Evans, Philip G. and Chatzidakis, Stylianos	Include	An empirical framework that boasts modularity and efficient performance over other simulation platforms. Models are probabilistic, focusing on effective means, distribution curves. What it also boasts is real time QRNG bit generation by using IBM's Qiskit platform, or ANU (Australian National University)	Wiley Advanced Physics Research	NuQKD
15	A Modeling Framework for Studying Quantum Key Distribution System Implementation Nonidealities	Journal	2014	Mailloux, Logan O. and Morris, Jeffrey D. and Grimaila, Michael R. and Hodson, Douglas D. and Jacques, David R. and Colombi, John M. and McLaughlin, Colin V. and Holes, Jennifer A.	Include	Dubbed as qkdX, is a simulation environment made using a C++ library OMNeT++. This paper is the base paper of [3] and defines various other components such as the framework of the environment and shows various components, like decoy state generator, classic pulse generator, optical pulse modulator, optical security layer, classical to quantum layer and etc. This paper explains most of the empirical formulations of components used. It is not a physical layer abstraction, but it shows the power of basic formulisms that can be used alongside physical layer abstraction	IEEE Access	qkdX
16	SimQN: A Network-Layer Simulator for the Quantum Network Investigation	Journal	2024	Chen, Lutong and Xue, Kaiping and Li, Jian and Yu, Nenghai and Li, Ruidong and Sun, Qibin and Lu, Jun	Exclude	A discrete-event driven network-layer simulation framework. Any more information is unavailable due to no paper access	IEEE Network	SimQN

17	Communication systems interactive software (comsis): modeling of components and its application to the simulation of optical communication systems	Journal	1998	Christine Le Brun and Eric Guillard and Jacques Citerne	Include	First ever communication simulation software designed on Fortran. It simulates the first proper laser in a component based environment rather than standalone. Some parts of other components such as optical fiber and amplifiers also are designed on the physical layer. Other components are modelled using the close approximation of Jones Calculus and uses Runge-Kutta method for simultaenous equation solvers. This paper closely matches our paper design, however design leverages newer computer hardware, which allows us to closely simulate on the physical abstraction layer	Applied Optics	LeBrun:98
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Gap Consideration: Among the simulation environments mentioned above, while many are component based, they are not implementing Quantum Protocols or Networks on the Physical layer, except few. Of the few that do, they do not present a noisy model for either the components or the quantum channel. Those that do more so focus on the network layer or direct implementation layer, which condenses most things down to closest approximations. The closest component based network/protocol simulation matching the work provided by our paper was done in 1998, after which researchers started to focus more towards bitstream communication. Additionally new simulation platforms provide more end results of network/protocol, focusing more on security, node based network infrastructure and less on the physical layer simulation. This might be due to the fact that many researchers in a race have opted to simulate the new Noise Intermediate-State Quantum (NISQ) era of quantum computing, which promises a breakthrough towards the fault-tolerance era of future quantum computing; however it is necessary to provide physical layer component simulation that can correlate information based on lab experiments to further strengthen understanding and underlying issues, instead of empirical formulations, or at the very least provide a combination of empirical and physical layer abstraction. Our design can be seen as an extension of work done in 1998, where we leverage modern computer power to simulate components on physical layer, while also using some empirical formulations. Combining both allows for a powerful tool that can be used to correlate lab experiments to simulations.

Possible Limitation(s) Consideration: During production of code and researching various components, no matter how powerful modern computers are, noise calculations take a long time to compute. Full physical layer abstraction can cause issues with Jones/Mueller Calculus representations (seen much recently with phase modulator implementation, which causes phase drift rather than polarization drift, which causes Jones Calculus to fail). We must deduce a spot where noise must be included, how physically complete the components are and how the simulation framework should be established, as all components must work on the same time and space. Additionally it must also be decided for how modular the code should be. High level of modularity has seem to cause some polynomial calculation issues which causes them to be possibly NP-Hard. Recent such example of NP-Hard formulism can be accounted towards wave-guide equations, which are extremely taxing to compute in a time and space format, while possible local minima convergent, or even non-converging convex solutions in presence of Noise. Many of such equations also dont have discretized matrix formulations.